US012406957B2

(12) **United States Patent**
Choi

(10) **Patent No.: US 12,406,957 B2**
(45) **Date of Patent: Sep. 2, 2025**

(54) **SPACERS FOR SEMICONDUCTOR DEVICE ASSEMBLIES**

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(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 521 days.

(21) Appl. No.: **17/890,834**

(22) Filed: **Aug. 18, 2022**

(65) **Prior Publication Data**

US 2024/0063166 A1 Feb. 22, 2024

(51) **Int. Cl.**
H01L 23/00 (2006.01)

(52) **U.S. Cl.**
CPC ***H01L 24/32*** (2013.01); ***H01L 24/16*** (2013.01); ***H01L 24/73*** (2013.01); *H01L 2224/16225* (2013.01); *H01L 2224/32225* (2013.01); *H01L 2224/73204* (2013.01)

(58) **Field of Classification Search**
CPC H01L 24/32; H01L 24/16; H01L 24/73
See application file for complete search history.

(56) **References Cited**

U.S. PATENT DOCUMENTS

2014/0110856 A1* 4/2014 Lin H01L 23/49816 257/774
2019/0164909 A1* 5/2019 Chang Chien H01L 24/19

* cited by examiner

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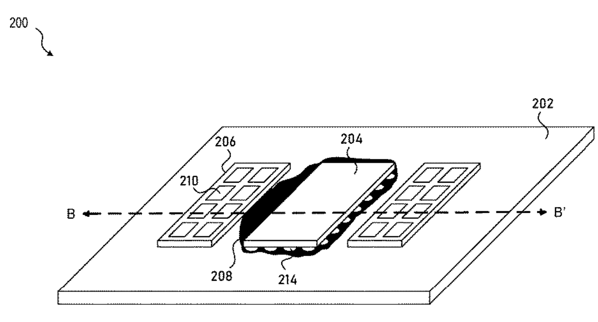
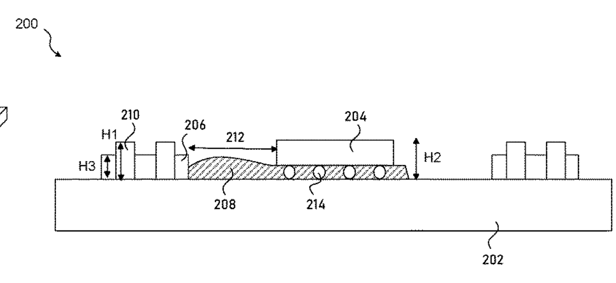
(57) **ABSTRACT**

A semiconductor device assembly including a substrate; a frame structure disposed on the substrate, the frame structure comprising peripheral walls defining an enclosed region; and a spacer disposed on the substrate within the enclosed region and restrained by the frame structure. A method of forming a semiconductor assembly, including forming a frame structure on a substrate, the frame structure comprising peripheral walls defining an enclosed region; attaching a semiconductor die to the substrate, the semiconductor die being adjacent to the frame structure; and dispensing a spacer on the substrate within the enclosed region, the spacer being restrained by the frame structure.

19 Claims, 8 Drawing Sheets





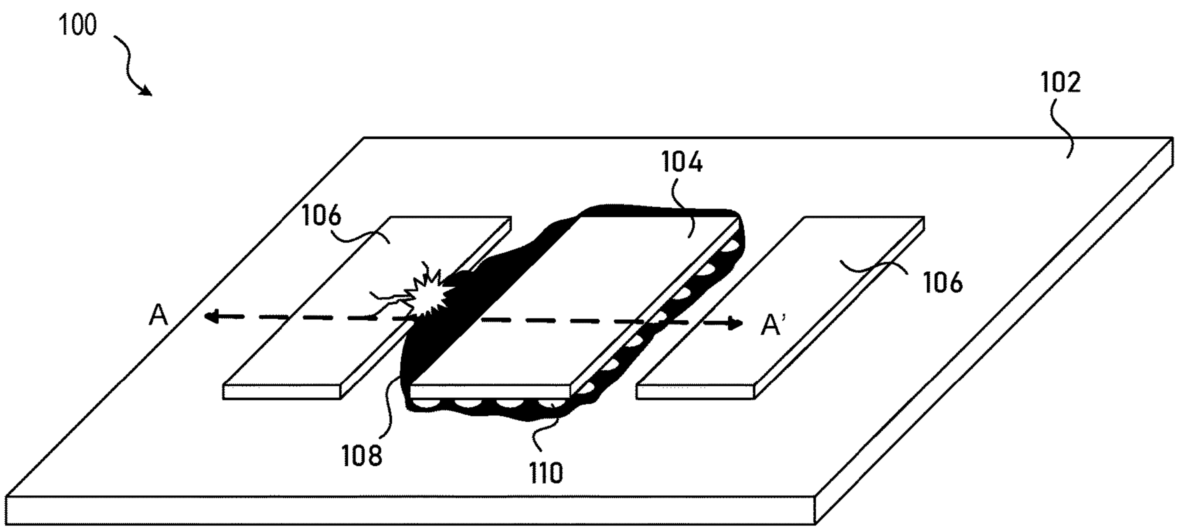


FIG. 1A

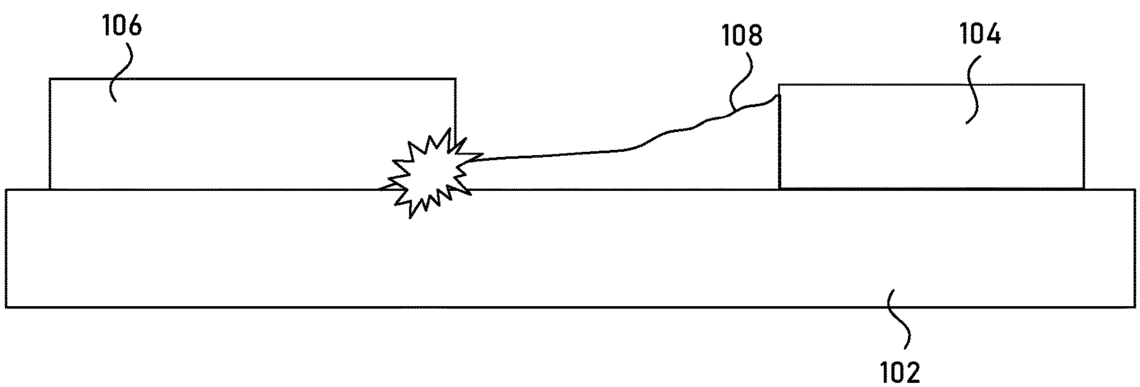


FIG. 1B

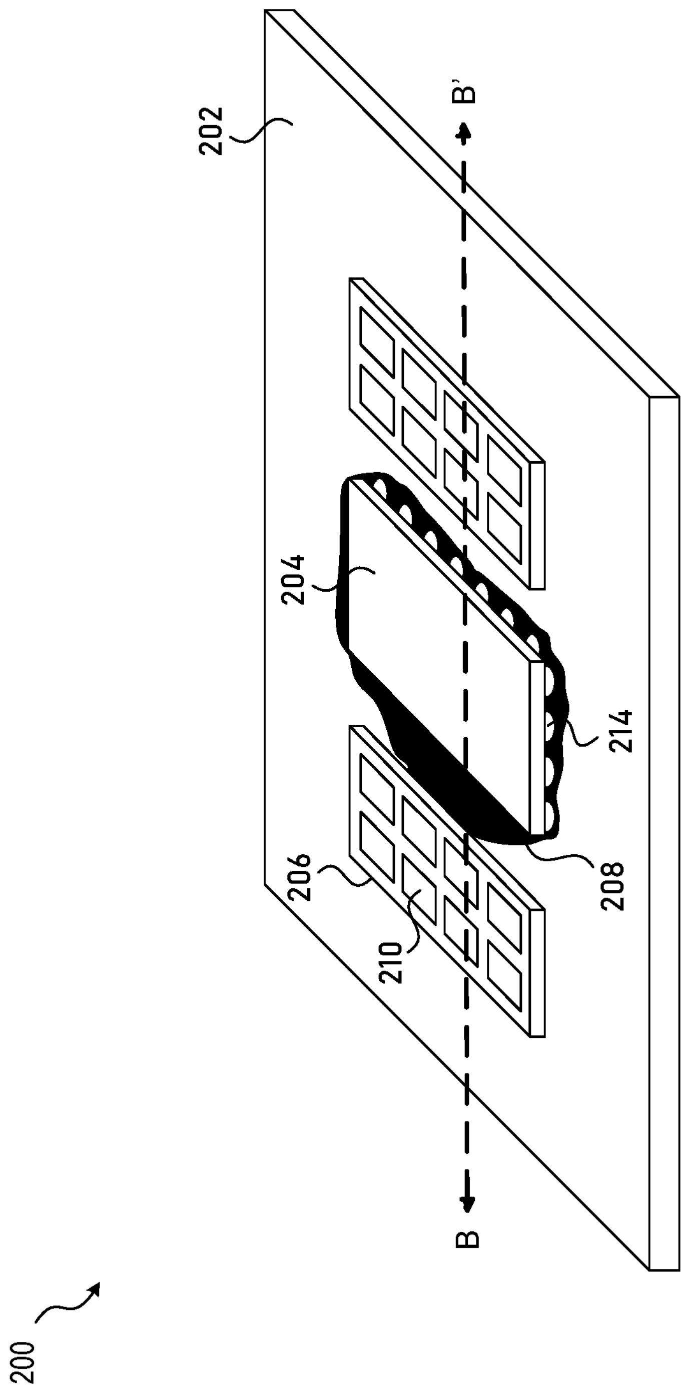


FIG. 2

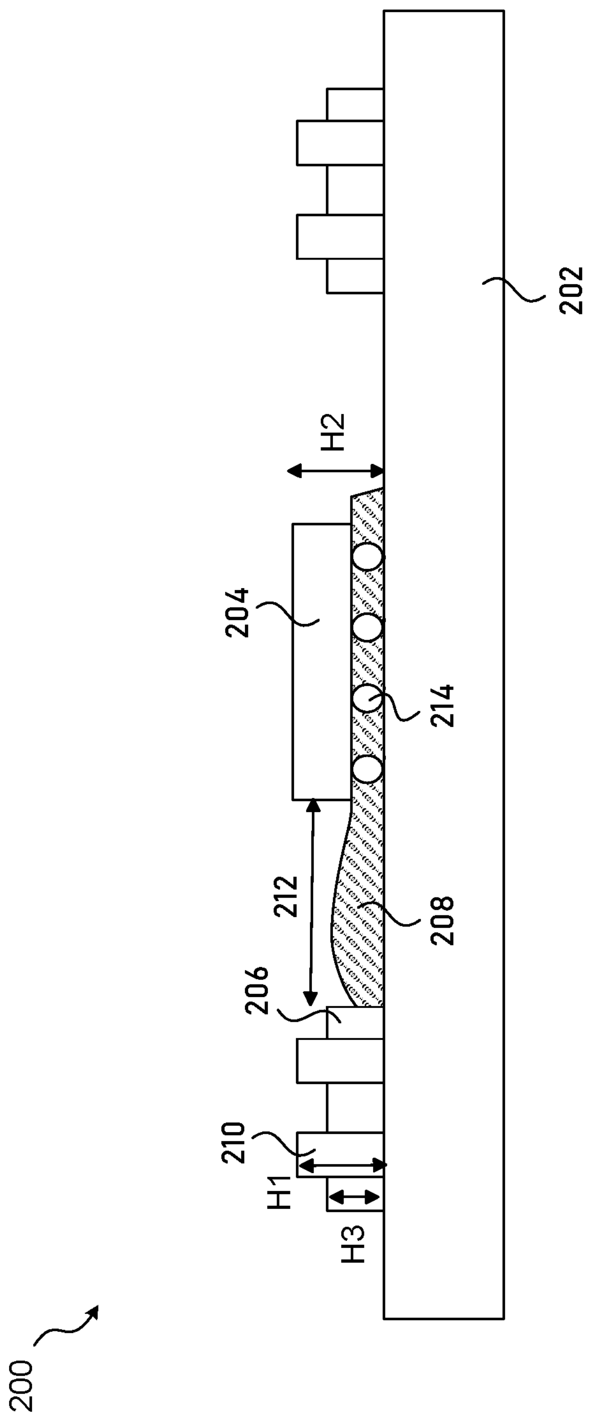


FIG. 3

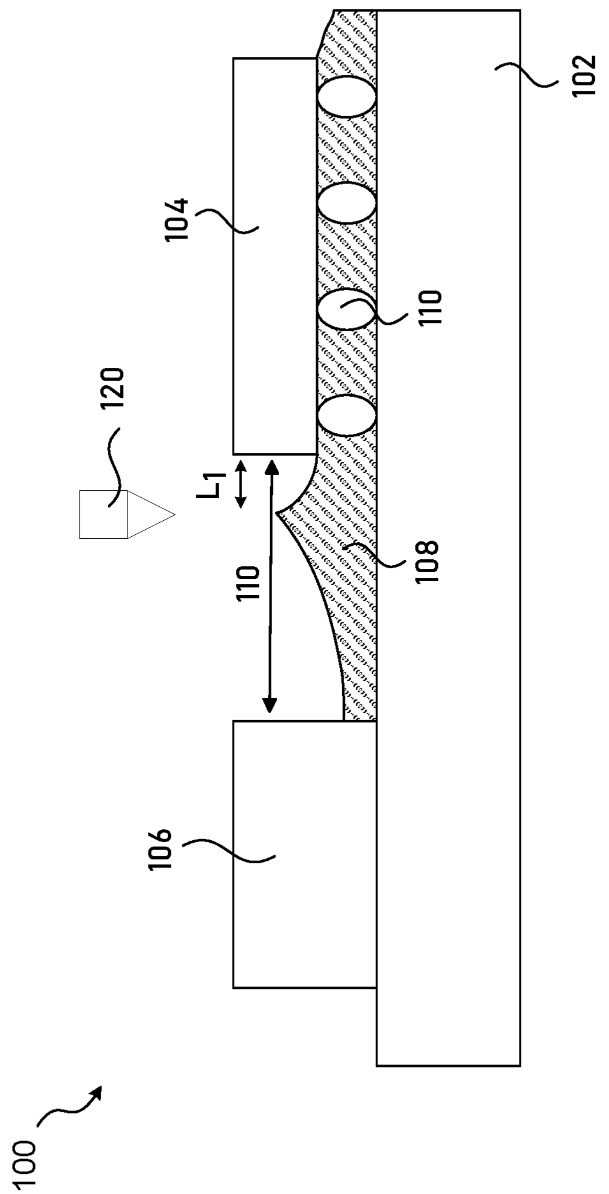


FIG. 4A

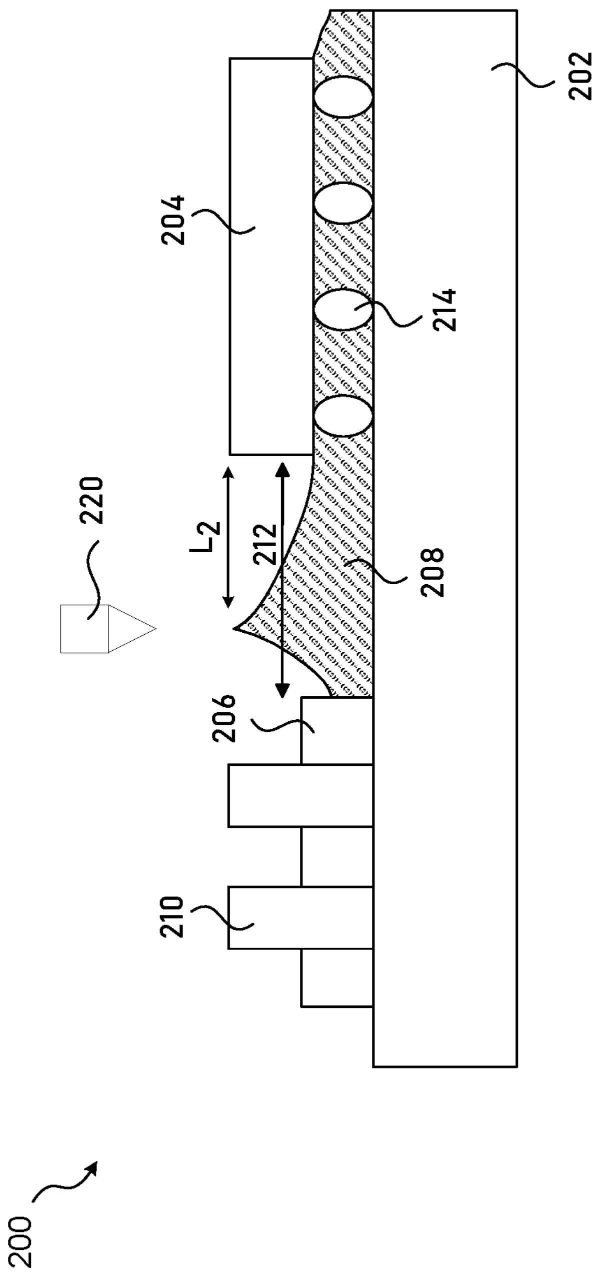


FIG. 4B

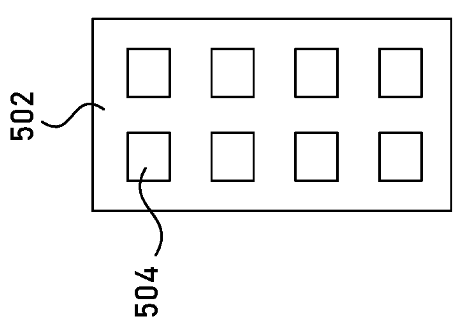


FIG. 5A

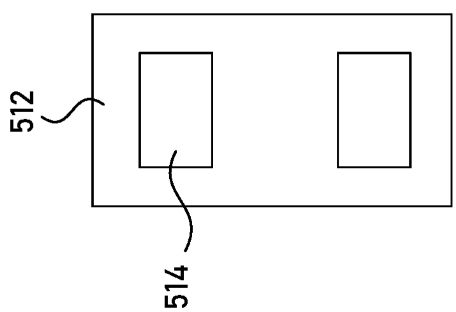


FIG. 5B

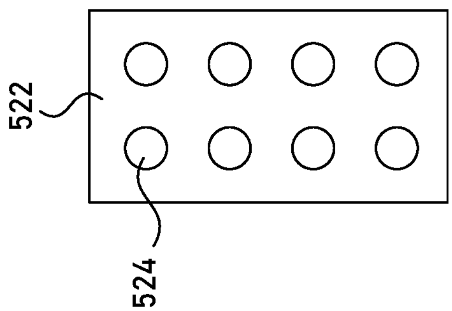


FIG. 5C

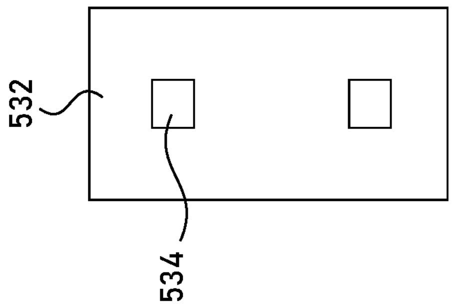


FIG. 5D

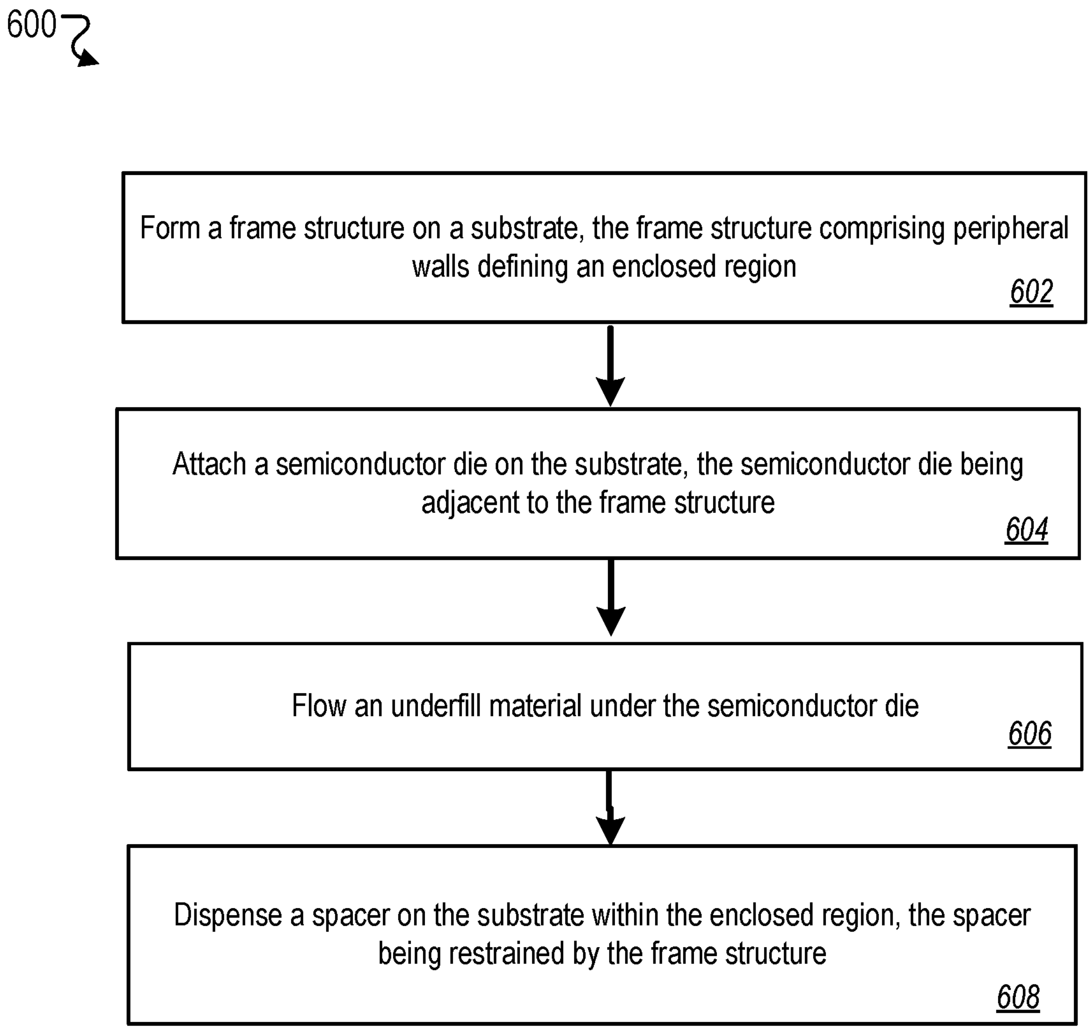

600
Form a frame structure on a substrate, the frame structure comprising peripheral walls defining an enclosed region
602
Attach a semiconductor die on the substrate, the semiconductor die being adjacent to the frame structure
604
Flow an underfill material under the semiconductor die
606
Dispense a spacer on the substrate within the enclosed region, the spacer being restrained by the frame structure
608

FIG. 6

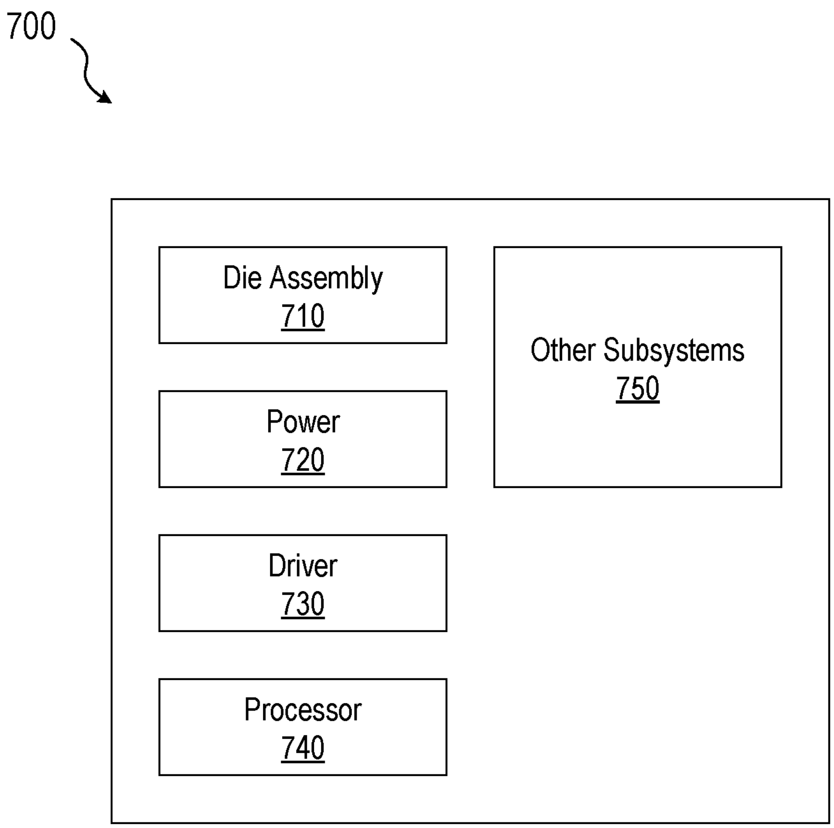

700
Die Assembly
710
Power
720
Driver
730
Processor
740
Other Subsystems
750

FIG. 7

SPACERS FOR SEMICONDUCTOR DEVICE ASSEMBLIES

TECHNICAL FIELD

The present disclosure generally relates to semiconductor devices, and more particularly relates to spacers for semiconductor device assemblies.

BACKGROUND

Semiconductor dies, including memory chips, microprocessor chips, logic chips, and imager chips are typically assembled by mounting a plurality of semiconductor dies, individually or in die stacks, on a substrate in a certain pattern. The assemblies can be used in mobile devices, computing, and/or automotive products. Spacers made of silicon can be used to support overhanging portions of large chips in the semiconductor device assemblies. Specifically, silicon spacers may help adhesively hold the semiconductor dies together prior to encapsulation and provide clearance for wire bonds loops between the semiconductor dies to the substrate or a high-level packaging. Although the silicon spacer fabrication is compatible with standard semiconductor device processes, it is sensitive to residues disposed on the substrate surface and could be cracked or delaminated during the semiconductor assembly processes.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. **1**A depicts a perspective view of a semiconductor device assembly including silicon spacers.

FIG. **1**B depicts a cross sectional view of a semiconductor device assembly including a silicon spacer.

FIG. **2** depicts a perspective view of a semiconductor device assembly including spacers according to embodiments of the present technology.

FIG. **3** depicts a cross sectional view of a semiconductor device assembly including spacers according to embodiments of the present technology.

FIG. **4**A depicts a cross sectional view of a semiconductor device assembly including a spacer and a semiconductor die.

FIG. **4**B depicts a cross sectional view of a semiconductor device assembly including a spacer and a semiconductor die according to embodiments of the present technology.

FIG. **5**A-**5**D depict planar views of spacers with various shapes and sizes for semiconductor device assembly according to embodiments of the present technology.

FIG. **6** is a flow chart illustrating a method of processing a semiconductor device assembly with spacers according to embodiments of the present technology.

FIG. **7** is a schematic view of a system that includes a semiconductor device configured according to embodiments of the presented technology.

The drawings illustrate only example embodiments and are therefore not to be considered limiting in scope. The elements and features shown in the drawings are not necessarily to scale, emphasis instead being placed upon clearly illustrating the principles of the example embodiments. Additionally, certain dimensions or placements may be exaggerated to help visually convey such principles. In the drawings, the same reference numerals used in different embodiments designate like or corresponding, but not necessarily identical, elements.

DETAILED DESCRIPTION

Conventional semiconductor assemblies utilize silicon spacers to support semiconductor components in a stacking structure prior to encapsulation, as well as providing clearance or standoff for wire bonding loops between the semiconductor components and a package substrate or other higher-level packaging. In addition, silicon spacers may be used in semiconductor device assemblies to improve packaging manufacturing processes yield. However, there is always a risk of delamination between the silicon spacer and the substrate. The spacer delamination may be caused by semiconductor die underfill material bleeding out under the silicon spacer. Further, foreign materials or residues introduced to spacer regions can also introduce additional mechanical stress on the spacers and lead to cracking thereon. An open circuit or other electrical failure may occur due to the spacer delamination and cracking, and the semiconductor assemblies can eventually fail to operate.

Various methods have been developed to improve the adhesion between the silicon spacer and substrate. For example, double side grinding has been conducted on both sides of the silicon spacer in order to increase surface roughness of the silicon spacer surfaces and adhesion thereon. In another approach, additional substrate cleaning processes has been applied to the semiconductor assemblies before attaching the silicon spacers. Those additional grinding or cleaning processes introduces additional fabrication steps and challenges, and therefore leads to a higher cost of silicon spacer processing for semiconductor device assemblies.

To address these challenges and others, the present technology applies frame structures to form spacers in semiconductor device assemblies. In particular, the present technology fabricates the frame structure on the substrate, the frame structure having peripheral walls and an enclosed region defined by the peripheral walls. Then spacer material is dispensed into the enclosed region of the frame structure to form spacers. The frame structure and the spacer can be made of solder resist and solder paste, respectively. In particular, the spacers can be partially embedded in the enclosed region of the frame structure and restrained by the frame structure. The spacers can have a height similar or higher than the frame structure to provide mechanical support on semiconductor dies disposes there above. Further, the frame structure can be disposed next to a semiconductor die and is configured to prevent underfill material bleeding out from the adjacent semiconductor die to the spacers. Moreover, the present technology can provide flexible sizes and locations for the spacers which allow different configurations of the semiconductor assemblies to accommodate various semiconductor die scaling requirement and additional semiconductor dies/die stacks configurations.

FIG. **1**A depicts a perspective view of a semiconductor device assembly **100** with silicon spacers **106**. The semiconductor device assembly **100** includes a substrate **102** and a semiconductor die **104** disposed on a top surface of the substrate **102**. In addition, the silicon spacers **106** are disposed on the top surface of the substrate **102** and adjacent to the semiconductor die **104**. As shown, the semiconductor die **104** is attached on the substrate **102** through solder bumps **110** and underfill adhesive material **108**.

The underfill adhesive material **108**, e.g., epoxy material, is commonly provided to fill the space between the bottom of the semiconductor die **104** and the top surface of the substrate **102** and between the solder bumps **110** to improve the semiconductor die **104** temperature cycling capability. In addition, the underfill adhesive material **108** is generally made of low viscosity epoxy material and is disposed around the perimeter of the semiconductor die **104**. The filling of epoxy materials under the semiconductor die **104** relies on capillary action to uniformly draw the epoxy material when it is still in liquid phase into the interior spaces between the semiconductor die **104** bottom surface and the substrate **102** top surface.

The silicon spacers **106** shown in FIG. **1**A can be cut or singulated silicon pieces, or other suitable materials, disposed on the substrate **102** and adjacent to the semiconductor die **104**. The silicon spacers **106** are usually configured to have a top surface equal to or slightly above a top surface of the semiconductor die **104** to provide mechanical support to additional semiconductor dies or die stacks disposed there above. The silicon spacers **106** can be secured on the substrate **102** by adhesive materials including adhesive past, adhesive tape, etc. The semiconductor device assembly **100** may include a b-state resin or solder paste to replace the silicon spacers **106** for similar applications.

One of the issues usually encountered during the semiconductor device assembly **100** is underfill epoxy bleed out which further lead to spacer defects. The underfill epoxy bleed out is a separation of certain formulation ingredients from the underfill epoxy material. Depending on the semiconductor die **104** solder paste formulations and substrate **102** morphology, the bleeding ingredients can also be solvents, low molecular weight resins, or additive adhesion promoters. The epoxy bleed out tends to occur on high energy surfaces such as silicon substrate without any organic coatings. Moreover, the epoxy bleed out can occur once the underfill epoxy material **108** is dispensed between the silicon spacers **106** and the semiconductor die **104**. Further, the epoxy bleed out can happen during thermal curing.

FIG. **1**B depicts a cross sectional view of the semiconductor device assembly **100** across the A-A' plane shown in FIG. **1**A. As shown, the underfill epoxy bleed out may cause the epoxy ingredients flow out of a keep out zone (KOZ) of the semiconductor die **104**. KOZ is a space identified around a semiconductor die attached on the substrate, within which no other semiconductor die or components are allowed in order to prevent cross contamination caused by under fill epoxy bleed out. As shown in FIG. **1**B, the underfill epoxy **108** may further bleed out to the silicon spacer region before attaching the silicon spacer **106** on the substrate **102**. The residue under fill epoxy disposed under the silicon spacer **106** may cause the silicon spacer delamination during or after the semiconductor device assembly **100** is completed. In addition, the silicon spacers **106** may be cracked during operating the semiconductor device assembly **100** because the underneath epoxy residue may cause additional mechanical stress thereon. Further, the underfill epoxy **108** may flow above the top surface of the semiconductor die **104** (not shown), when the epoxy dispense location is configured to be close to the edge of the semiconductor die **104** to prevent underfill epoxy bleed out to the spacer region, causing a underfill material creeping defect. As microelectronics continue to move towards smaller form factor, dimensions of the semiconductor die and the KOZ are scaled, bringing in more challenges in semiconductor device assembly. It becomes increasingly critical to control the underfill epoxy bleed out to prevent damages to the spacers and reduce defects.

FIG. **2** depicts a perspective view of a semiconductor device assembly **200** including spacers **210** according to embodiments of the present technology. As shown, the semiconductor assembly **200** include a substrate **202**, a semiconductor die **204**, and a plurality of spacers **210**. The substrate **202** can be electrically coupled to the semiconductor die **204** and additional semiconductor dies (not shown) disposed above or adjacent to the semiconductor die **204**, such that the semiconductor die **204** interfaces between the additional semiconductor dies and a host (e.g., a host processor of an electronic device) in communication with the semiconductor assembly **200**. In this scenario, the semiconductor assembly has a relatively small footprint for high density semiconductor packaging, e.g., high bandwidth memory (HBM) applications.

The semiconductor device assembly **200** also includes a frame structure **206** within which the plurality of spacers are disposed. Specifically, the frame structure **206** has peripheral walls defining an enclosed region (e.g., a laterally-enclosed region open to the top) and is disposed adjacent to the semiconductor die **204**. The plurality of spacers **210** are disposed on the substrate **202** and within the enclosed region of the frame structure **206**. In one embodiment and as shown in FIG. **2**, the semiconductor device assembly **200** includes 2 frame structures **206** each being disposed next to a longer edge of the semiconductor die **204**. Each of the frame structures **206** has trench openings aligned in columns and rows. In particular, each of the frame structures **206** includes 8 enclosed regions, i.e., trench openings arranged in a 2×4 array. Further, each of the frame structures **206** includes 8 spacer pillars, each of the spacer pillars being disposed and restrained in the trench openings of the frame structures **206**.

In some embodiments, the plurality of spacers **210** may be partially restrained in the frame structure **206**. For example, the frame structure **206** may include a plurality of enclosures defined by its peripheral walls. The plurality of spacers **210** can be partially disposed within the plurality of enclosures, respectively. In this example, the plurality of spacers **206** may be taller than the frame structure **206** on the substrate **202**.

In some embodiments, the substrate **202** may include, for example, a printed circuit board, a multimedia card substrate, or other suitable interposer having electrical connectors, such as metal traces, vias, or other suitable connectors. In various embodiment, the substrate **202** can be made of silicon, silicon-on-insulator, compound semiconductor (e.g., Gallium Nitride), or other suitable substrates and can have any of variety of integrated circuit components or functional features for managing memory or other components included in the semiconductor device assembly **200**.

In some embodiments, the semiconductor device assembly **200** can be a memory device package and the semiconductor die **204** can be a controller die that is positioned under memory dies and configured to provide memory management for the memory package. In some embodiments, memory devices can be flash memory (e.g., eMMC memory, Universal Flash Storage, etc.) with multi-die memory packages suitable for mobile devices (e.g., smart phones, tablets, MP3 players, etc.), digital cameras, routers, gaming systems, navigation systems, computers, and other consumer electronic devices. For example, the multi-die memory packages can be, for example, flash memory packages, such as NAND packages, NOR packages, etc. The semiconductor die **204** can handle memory management on the semiconductor device assembly **200** so that a host processor is free to perform other tasks. In various embodiments, the semiconductor die **204** can include circuitry, software, firmware, memory, or combinations thereof.

In some embodiments, the semiconductor die **204** can be attached to the substrate **202** by a plurality of solder bumps **214** and an underfill adhesive material **208**. The underfill adhesive material **208** can be any adhesive materials (e.g., epoxy resin, adhesive paste, etc.), an adhesive laminate (e.g., adhesive tape, die-attach or dicing die-attach film, etc.), or other suitable materials. As shown in FIG. **2**, the underfill adhesive material **208** may bleed out and is further disposed in a dispense region between the semiconductor die **204** and adjacent frame structure **206**. The underfill adhesive material **208** may have a greater thickness outside the outer edges of the semiconductor die **204** than under the semiconductor die **204** bottom surface, so as to accommodate the plurality of solder bumps **214** between the semiconductor die **204** and the substrate **202**.

In some embodiments, the frame structure **206** can be made of solder resist. Solder resist is a permanent coating of a resin formulation and can be processed by a liquid photo-imagable (LPI) technique. Here, the solder resist used in the present technology can be in the form of a liquid photo-polymer, epoxy, or epoxy-acrylate resin. The solder resist thickness is typically ranging from 10 um to 50 um with 5 um variations. Once dry after dispensing the solder resist on substrate **202**, it is exposed to an image pattern required and then developed to reveal the required solder resist pattern. After developing, the solder resist is further heat cured to ensure a tough durable finish. In some other embodiments, the frame structure **206** may be made of insulating inks.

The frame structure **206**, in the present technology, is configured to restrain the spacers **210** in the required regions. For example, the plurality of spacer pillars **210** are partially embedded in the patterns of the frame structure **206** and are disposed within the spherical walls of the frame structure **206**. In addition, the frame structure **206** can prevent adhesive underfill material **208**, e.g., underfill epoxy, from bleeding out and dispensed under the spacer **210**. When the spacer **210** is being dispensed on the substrate **202**, in this example, the frame structure **206** works as a permanent protective dam, protecting the spacer **210** from the underfill adhesive material **208**.

In some embodiments, solder resists such as alkaline developable solder resist, UV curable solder resist, and/or thermally curable solder resist can be used to construct the frame structure **206** for the semiconductor device assembly **200**. For example, alkaline developable solder resist can be implemented to shape a precise pattern onto the substrate **202** by exposing a negative film with the spacers **210** pattern via a screen print, spray coat, and/or curtain coat. Alternatively, the UV curable solder resist can be applied on the substrate **202** and be hardened under UV light for pattern printing via a screen print method. In addition, thermally curable solder resist can be utilized in the present technology through hardening the solder resist by applying heat.

Once the frame structure **206** is formed on the substrate **202**, spacers **210** can be processed within the frame structure **206**. In some embodiments, the spacers **210** can be made of solder pastes, e.g., a combination of metal solder particles and sticky flux with consistency of putty. The flux ingredient of the solder paste not only cleans the soldering surfaces of impurities and oxidation, but also provides an adhesive that holds the solder paste on the substrate **202**. In modern technologies, the metal solder particles of the solder paste may be made of lead-free alloys.

In some embodiments, the spacer **210** may be dispensed to the frame structure **206** by a stencil-printing process. For example, solder paste can be dispensed into the patterns of the frame structure **206** by a solder paste printer. Alternatively, the solder paste may be dispensed pneumatically by jet printing, wherein the solder paste is ejected onto the substrate **202** through one or more nozzles. The printing of spacers **210** may followed by a reflow soldering process. A suitable reflow temperature may be selected to reflow the solder paste to suit them in the pattern of the frame structure **206**. For example, the solder paste may be formed into a pillar shape with rectangular spherical edges in the horizontal plane, similar to the trench opening pattern of the frame structure **206**, after the reflowing. Further, a reasonably rapid cool-down period may be required to solidify the spacers **210**.

As shown in FIG. **2**, the semiconductor device assembly **200** includes two frame structures and a plurality of spacers **210** disposed adjacent to both long edges of the semiconductor die **204**. In some other embodiments, there may be additional frame structures and spacers embedded therein that are disposed on the substrate **202** and adjacent to the semiconductor **204**. For example, the semiconductor device assembly **200** may also include frame structures adjacent to the shorter edges of the semiconductor die **204**. In various embodiments, the semiconductor device assembly **200** may include multiple frame structures and spacers embedded therein, the multiple frame structures being adjacent to each edge of the semiconductor die **204**.

Now turning to FIG. **3** which depicts a cross sectional view of the semiconductor device assembly **200** across the B-B' plane according to embodiments of the present technology. As described in FIG. **2**, the semiconductor device assembly **200** may include two frame structures each disposed adjacent to the longer edge of the semiconductor die **204**. In some embodiments, the plurality of spacers **210** are at least partially embedded in the frame structures **206** and in a shape of pillars. Specifically, each of the plurality of spacers **210** is in contact with the walls of the trench openings of the frame structures **206** and protruding upward. The semiconductor die **204** may be a controller flip chip die mounted to the substrate **202** via a plurality of solder bumps **214** and underfill adhesive material **208**. The underfill material **208** may be an epoxy polymer such as aluminum-oxide, silica, or other know underfill materials. The semiconductor die **204** can perform various functionalities and is referred to herein as flip chip based on its connection to the substrate **202**. Although the semiconductor die **204** is shown as a flip chip, other types of devices and chips can be used with the spacer **210** in the present technology. Here, the solder bumps **214** may spread between a bottom surface of the semiconductor die **204** and the substrate **202** for interconnections. In some other embodiments, the solder bumps **214** may be only located near outer edges of the semiconductor die **204**, although are not so limited.

In some embodiments, additional semiconductor devices (not shown) may be mounted over the semiconductor devices shown in FIG. **3**. The additional semiconductor devices can be an active die, such as a non-volatile storage memory such as NAND, a dynamic random-access memory (DRAM), or other memory chips, microprocessor chip, logic chip, or imaging chip as a bottom die in a die stack. The additional semiconductor devices can be mounted on the semiconductor die **204** and at least partially carried by the spacers **210** because of its relatively large size or configurations within the semiconductor assembly **200**. For example, the additional semiconductor devices may overhang a portion of the semiconductor die **204** and a portion of the spacers **210**. In various embodiments, the additional semiconductor devices may have a relatively large dimension, completely covering the top surface of the semiconductor die **204** and overhanging at least a portion of the spacers **210**.

In some embodiments, other dies, die stacks, and/or components can be mounted to the substrate **202**. It should also be understood that the spacer **210** and frame structure **206** can be disposed above any dies, die stacks, and/or components of the semiconductor assembly. For example, the spacer **210** and the frame structure **206** can be mounted to a semiconductor die disposed on the substrate **202** and the top surface of the spacer **210** can be leveled to a top surface of the semiconductor die **204**, before further mounting any additional semiconductor dies/die stacks there above.

As shown in FIG. **3**, the plurality of spacers **210** has a height H**1** from the mounting surface of the substrate **202** to a top surface of the spacer **210**. The spacer height H**1** may be substantially the same as a height H**2** of the peripheral walls of the frame structure **206** marked from the mounting surface of the substrate **202** to the top surface of the semiconductor die **204**, within a tolerance. As a result, when the additional semiconductor devices are mounted over the semiconductor die **204** and the spacer **210**, the additional semiconductor devices are leveled and there is no open space or gap created between the additional semiconductor devices and the top surfaces of the spacers **210** and the semiconductor die **204**. As described, the spacers **210** are at least partially embedded in the frame structure **206** which is configured to constrain the spacer material such as solder paste. The height H**3** of the frame structure **206** may be equal to or lower than the height H**1** of the spacers **210**. In some embodiments, the height H**1** of the spacers may range from 50 um to 100 um. The height H**3** of the frame structure **206** may range from 10 um to 50 um.

As described, the frame structure **206** may include a pattern of enclosed regions, e.g., trench opening arrays. The spacer materials, e.g., solder paste, can be dispensed into the trench opening arrays of the frame structure **206**. Specifically, the spacer material may fulfill the trench openings. While the dispensing continues, the spacer material may further overstep the top surface of the trench openings/frame structure **206** and reveal a taller pillar structure. The spacer pillars may well maintain a rectangular spherical shape from the trench openings until reaching the height H**1** and forming a planar top surface due to its high viscosity.

In some embodiments, the underfill adhesive material **208** can be dispensed into the KOZ **212** between the frame structure **206** and the adjacent edge of the semiconductor die **204**. The KOZ **212** may have a length ranging from 500 um to 1 mm. As described, the underfill adhesive material **208** can be firstly dispensed on the substrate **202** in liquid phase and then uniformly drawn into the interior spaces between the semiconductor die **204** bottom surface and the substrate **202** top surface based on capillary action. Even though the dispensing location can be configured to be closer to the edge of the semiconductor **204**, the underfill material **208** may inevitably bleed out towards the spacers **210**. In the present technology, the underfill material bleed out stops at the frame structure **206**, which is disposed between the underfill material **208** and the spacers **210**. Here, the frame structure **206** performs as a mechanical barrier or a dam to reduce the underfill material bleed out into the region of the spacers **210**, as shown in FIG. **3**.

FIG. **4**A depicts a cross sectional view of the semiconductor device assembly **100** with the silicon spacer **106** and the semiconductor die **104**. This semiconductor device assembly **100** also includes a KOZ **110** around one edge of the semiconductor die **104**. Generally, the underfill epoxy material **108** is dispensed from a nozzle **120** disposed above the KOZ **110** region and drawn into the interior space between the semiconductor die **104** and the substrate **102**. To improve the underfill efficiency, the dispense location **1**/nozzle position is usually configured to be closer to the edge of the semiconductor die **104**. The dispense location **1** may have a distance L**1** to the adjacent edge of the semiconductor die **104**. The distance L**1** may be ranging from 20 um to 200 um.

This configuration can be also implemented for semiconductor device assemblies including traditional silicon spacers to reduce epoxy material bleed out by keeping the dispense location further away from adjacent silicon spacers, as shown in FIG. **4**A. Along with semiconductor device scaling, however, this configuration is getting more challenging. First, the KOZ **110** length may be reduced to enhance the density of semiconductor dies within the semiconductor assemblies. The reduced length of KOZ **100** inevitably increases the risk of underfill epoxy **108** bleeding out towards to the silicon spacer **106**, causing more silicon spacer cracking or delamination defects. Second, the thickness of the semiconductor die **104** may be reduced when scaling the semiconductor dies. A thinner/lower top surface of the semiconductor die **104** may increase a risk of underfill adhesive material **108** creeping defects, i.e., the underfill epoxy flow on the top surface of the semiconductor die **104**. The creeping defect may range up to 200 um from an edge to a center above the top surface of the semiconductor die **104**. In addition, the underfill epoxy flow along the adjacent edge of the semiconductor die **104** is expected to be faster to the underneath region and its opposite end edge, therefore causing an accumulation of underfill epoxy at the adjacent edge of the semiconductor die **104** and increasing the risk of creeping defect.

FIG. **4**B depicts a cross sectional view of the semiconductor device assembly **200** according to the present technology. As described, the spacer **210** can be restrained in the frame structure **206**, both the spacers **210** and the frame structure **206** being adjacent to the semiconductor die **204**. The underfill adhesive material **208** can be dispensed onto the substrate **202** through the KOZ **212** and then drawn into the interior space between the semiconductor die **204** bottom surface and the substrate **202** top surface.

In some embodiments and as shown in FIG. **4**B, the underfill adhesive material **208** can be dispensed on the substrate **202** from a nozzle **220** disposed above the KOZ **212**. For example, the dispense location **2**/nozzle **220** may have a distance L**2** to the adjacent edge of the semiconductor die **204**. The L**2** may be larger than the distance L**1** described in semiconductor device assembly **100**. Here, the distance L**2** may be ranging from 175 um to 400 um. In this example, the frame structure **206** restrains the spacers **210** and surrounds the spacers **210**. As shown in FIG. **4**B, the frame structure **206** performs as a dam to prevent underfill material **208** bleeding out into the spacer regions. Here, adjusting the underfill material dispensing location/nozzle **220** position to be closer to the frame structure will not cause any degradation on the spacers **210** because the frame structure **206** can effectively isolate the underfill material **208** from the spacers **210**. In addition, the adjusted dispense location **2** further away from the adjacent edge of the semiconductor die **204** may help reduce the risk of underfill material creeping defects. For example, by adjusting the underfill material **208** dispense location closer to the edge of the frame structure **206**, the underfill material **208** can be less accumulated at the adjacent edge of the semiconductor die **204**. Therefore, the risk of creeping the underfill material **208** too far along the top surface of the semiconductor die **204** can be effectively reduced.

An expected advantage and benefit of the spacers **210** in the present technology are the ability to design the size (e.g., length, width), shapes, thickness, and location of the spacers **210** as well as the frame structures **206** within the semiconductor assembly 200. FIGS. 5A-5D depict planar views of spacers restrained in peripheral walls of frame structures with various shapes and sizes for semiconductor device assembly according to embodiments of the present technology.

As described, a plurality of spacers can be restrained in a frame structure. For example, spacers **504** can be arranged in columns and rows within the frame structure **502**. As shown in FIG. **5**A, the spacer pillars **504** can be arranged in a 2×4 array and disposed in the frame structure **502**. Each of the spacer pillars **504** is in a square shape with an edge length ranging from 50 um to 500 um. Alternatively, as shown in FIG. **5**B, spacers can be arranged only in a column within a frame structure. For example, two spacer pillars **514** can be arranged vertically within the frame structure **512**. Each of these spacer pillars **514** may have a larger size than the spacer **504**, e.g., having an edge length ranging from 100 um to 1 mm.

The spacers for the semiconductor device assembly in the present technology can be in different shapes. The shape of the spacers may be originated from the enclosed regions formed within the frame structures. For example, as shown in FIG. **5**C, the spacer pillars **524** each have a circular shape in the horizontal plane. In addition, the spacer pillars **524** are arranged in a 2×4 array within the frame structure **522**. Each of the spacer pillars **524** may have a similar dimension to the spacer pillars **504**, e.g., having a radius ranging from 50 um to 500 um.

The semiconductor device assembly may include smaller size spacers to save the fabrication cost. For example, as shown in FIG. **5**D, two spacer pillars **534** can be arranged in a column and restrained in the frame structure **532**. Each of the spacer pillars **534** may be in a square shape and have a similar size to the spacer pillars **504**, e.g., having an edge length ranging from 50 um to 500 um. The frame structures shown in FIGS. **5**A-**5**D are all in a rectangular peripheral shape. In various embodiments, the frame structures' shape can be various. For example, the frame structure may be in a square peripheral shape, a circular shape, or a polygon shape.

This flexible sizing and shapes of the spacers allow for different configurations of the semiconductor assemblies to accommodate different semiconductor die scaling requirement and different additional semiconductor dies/die stacks configurations. For example, for a HBM assembly including a number of stacked memory dies disposed above the spacers, a configuration of a smaller number of spacers each having a larger size may be implemented to reduce the mechanical stress applied on each of the spacers. Alternatively, the HBM assembly may utilize a large number of spacers each having a relative smaller size for the same purpose. In another example, for semiconductor device assembly in which mechanical stress is not a concern, a smaller number of spacers each having a smaller size may be a good fit.

Turning now to FIG. **6**, a flow chart illustrating a method **600** of forming spacers for semiconductor device assemblies. The method **600** includes forming a frame structure on a substrate, the frame structure comprising peripheral walls defining an enclosed region, at **602**. For example, the frames structures **206** can be fabricated on the substrate **202** and adjacent to the semiconductor die **204**. The frame structures **206** can be made by solder resist and dispensed by the LPI technique. In addition, the frame structure **206** may include peripheral walls that define enclosed regions, as shown in FIG. **2**.

The method **600** also includes attaching a semiconductor die on the substrate, the semiconductor die being adjacent to the frame structure, at **604**. For example, the semiconductor die **204** can be attached on the substrate **202** by a plurality of solder bumps **214**. The semiconductor die **204** being placed adjacent to the frame structure **206**.

Further, the method **600** includes flowing an underfill material under the semiconductor die, at **606**. For example, adhesive underfill material **208** can be dispensed into the KOZ **212** between the frame structure **206** and the adjacent edge of the semiconductor die **204**. The underfill adhesive material **208** can be firstly dispensed on the substrate **202** in liquid phase and then uniformly drawn into the interior spaces between the semiconductor die **204** bottom surface and the substrate **202** top surface based on capillary action.

Lastly, the method **600** includes dispensing a spacer on the substrate within the enclosed region, the spacer being restrained by the frame structure, at **608**. For example, the spacer material **210** can be dispensed in the enclosed region of the frame structure **206**. The spacer pillars **210** can be made of solder paste and dispensed by a stencil-printing process. Moreover, the spacer pillars **210** can be partially embedded in the frame structure **206** and have a planar top surface, as shown in FIG. **3**.

Any one of the semiconductor structures described above with reference to FIGS. **2**-**5**D can be incorporated into any of a myriad of larger and/or more complex systems, a representative example of which is system **700** shown schematically in FIG. **7**. The system **700** can include a semiconductor device **710**, a power source **720**, a driver **730**, a processor **740**, and/or other subsystems or components **750**. The semiconductor device **710** can include features generally similar to those of the semiconductor devices described above and can therefore include spacers and frame structures described in the present technology. The resulting system **700** can perform any of a wide variety of functions, such as memory storage, data processing, and/or other suitable functions. Accordingly, representative systems **700** can include, without limitation, hand-held devices (e.g., mobile phones, tablets, digital readers, and digital audio players), computers, and appliances. Components of the system **700** may be housed in a single unit or distributed over multiple, interconnected units (e.g., through a communications network). The components of the system **700** can also include remote devices and any of a wide variety of computer-readable media.

Specific details of several embodiments of semiconductor devices, and associated systems and methods, are described below. A person skilled in the relevant art will recognize that suitable stages of the methods described herein can be performed at the wafer level or at the die level. Therefore, depending upon the context in which it is used, the term "substrate" can refer to a wafer-level substrate or to a singulated, die-level substrate. Furthermore, unless the context indicates otherwise, structures disclosed herein can be formed using conventional semiconductor-manufacturing techniques. Materials can be deposited, for example, using chemical vapor deposition, physical vapor deposition, atomic layer deposition, plating, electroless plating, spin coating, and/or other suitable techniques. Similarly, materials can be removed, for example, using plasma etching, wet etching, chemical-mechanical planarization, or other suitable techniques.

In accordance with one aspect of the present disclosure, the semiconductor devices illustrated above could be memory dice, such as dynamic random access memory (DRAM) dice, NOT-AND (NAND) memory dice, NOT-OR

(NOR) memory dice, magnetic random access memory (MRAM) dice, phase change memory (PCM) dice, ferroelectric random access memory (Fe RAM) dice, static random access memory (SRAM) dice, or the like. In an embodiment in which multiple dice are provided in a single assembly, the semiconductor devices could be memory dice of a same kind (e.g., both NAND, both DRAM, etc.) or memory dice of different kinds (e.g., one DRAM and one NAND, etc.). In accordance with another aspect of the present disclosure, the semiconductor dice of the assemblies illustrated and described above could be logic dice (e.g., controller dice, processor dice, etc.), or a mix of logic and memory dice (e.g., a memory controller die and a memory die controlled thereby).

The devices discussed herein, including a memory device, may be formed on a semiconductor substrate or die, such as silicon, germanium, silicon-germanium alloy, gallium arsenide, gallium nitride, etc. In some cases, the substrate is a semiconductor wafer. In other cases, the substrate may be a silicon-on-insulator (SOI) substrate, such as silicon-on-glass (SOG) or silicon-on-sapphire (SOP), or epitaxial layers of semiconductor materials on another substrate. The conductivity of the substrate, or sub-regions of the substrate, may be controlled through doping using various chemical species including, but not limited to, phosphorous, boron, or arsenic. Doping may be performed during the initial formation or growth of the substrate, by ion-implantation, or by any other doping means.

The functions described herein may be implemented in hardware, software executed by a processor, firmware, or any combination thereof. Other examples and implementations are within the scope of the disclosure and appended claims. Features implementing functions may also be physically located at various positions, including being distributed such that portions of functions are implemented at different physical locations.

As used herein, including in the claims, "or" as used in a list of items (for example, a list of items prefaced by a phrase such as "at least one of" or "one or more of") indicates an inclusive list such that, for example, a list of at least one of A, B, or C means A or B or C or AB or AC or BC or ABC (i.e., A and B and C). Also, as used herein, the phrase "based on" shall not be construed as a reference to a closed set of conditions. For example, an exemplary step that is described as "based on condition A" may be based on both a condition A and a condition B without departing from the scope of the present disclosure. In other words, as used herein, the phrase "based on" shall be construed in the same manner as the phrase "based at least in part on."

As used herein, the terms "top," "bottom," "over," "under," "above," and "below" can refer to relative directions or positions of features in the semiconductor devices in view of the orientation shown in the Figures. These terms, however, should be construed broadly to include semiconductor devices having other orientations, such as inverted or inclined orientations where top/bottom, over/under, above/below, up/down, and left/right can be interchanged depending on the orientation.

It should be noted that the methods described above describe possible implementations, and that the operations and the steps may be rearranged or otherwise modified and that other implementations are possible. Furthermore, embodiments from two or more of the methods may be combined.

From the foregoing, it will be appreciated that specific embodiments of the invention have been described herein for purposes of illustration, but that various modifications may be made without deviating from the scope of the invention. Rather, in the foregoing description, numerous specific details are discussed to provide a thorough and enabling description for embodiments of the present technology. One skilled in the relevant art, however, will recognize that the disclosure can be practiced without one or more of the specific details. In other instances, well-known structures or operations often associated with memory systems and devices are not shown, or are not described in detail, to avoid obscuring other aspects of the technology. In general, it should be understood that various other devices, systems, and methods in addition to those specific embodiments disclosed herein may be within the scope of the present technology.

What is claimed is:

1. A semiconductor device assembly, comprising:

a substrate;

a frame structure disposed on the substrate, the frame structure comprising peripheral walls defining an enclosed region;

a spacer disposed on the substrate within the enclosed region and restrained by the frame structure, the spacer being nonconductive;

a semiconductor die disposed on the substrate and disposed adjacent to the frame structure and the spacer; and

an underfill material layer that is in contact with the peripheral walls of the frame structure and that extends under the semiconductor die.

2. The semiconductor device assembly of claim **1**, wherein the frame structure comprises solder resist, and wherein the spacer comprises solder paste.

3. The semiconductor device assembly of claim **1**, wherein the frame structure is configured to prevent the underfill material layer from horizontally dispensing under the spacer disposed on the substrate.

4. The semiconductor device assembly of claim **1**, wherein a dispense space exists between the semiconductor die and the adjacent frame structure, and wherein the dispense space has a length between the semiconductor die and the adjacent frame structure, the dispense space length ranging from 500 um to 1 mm.

5. The semiconductor device assembly of claim **4**, wherein the underfill material layer is disposed in the dispense space.

6. The semiconductor device assembly of claim **1**, wherein the spacer has a planar top surface.

7. The semiconductor device assembly of claim **6**, wherein the top surface of the spacer is equal to or higher than a top surface of the frame structure.

8. The semiconductor device assembly of claim **7**, wherein the frame structure has a height ranging from 10 um to 50 um, and wherein the spacer has a height ranging from 50 um to 100 um.

9. The semiconductor device assembly of claim **1**, wherein the peripheral walls define a plurality of enclosures including the enclosure, and wherein the spacer comprises a plurality of pillars that are at least partially restrained in corresponding ones of the plurality of enclosures of the frame structure.

10. The semiconductor device assembly of claim **9**, wherein the plurality of pillars have at least one of a rectangular peripheral shape, a square peripheral shape, a circular peripheral shape, or a polygon peripheral shape in a horizontal plane.

11. The semiconductor device assembly of claim **1**, wherein the frame structure has at least one of a rectangular peripheral shape or a square peripheral shape in a horizontal plane.

12. The semiconductor device assembly of claim **1**, further comprising a plurality of memory dies, the memory dies being disposed above the semiconductor die and at least partially carried by the spacer.

13. A substrate for a semiconductor device assembly, the substrate comprising:

a first surface including a plurality of die contacts;

a second surface opposite the first surface including a plurality of external contacts operably coupled to the plurality of die contacts;

a frame structure at the first surface, the frame structure comprising peripheral walls defining an enclosed region exclusive of the plurality of die contacts and configured to restrain a spacer; and

an underfill material layer that is in contact with the peripheral walls of the frame structure and that extends through the plurality of die contacts.

14. The substrate of claim **13**, wherein the frame structure comprises solder resist, and wherein the spacer comprises solder paste.

15. The substrate of claim **13**, wherein the frame structure is configured to prevent the underfill material layer from horizontally dispensing into the enclosed region.

16. A method of forming a semiconductor assembly, comprising:

forming a frame structure on a substrate, the frame structure comprising peripheral walls defining an enclosed region;

attaching a semiconductor die to the substrate, the semiconductor die being adjacent to the frame structure;

dispensing a spacer on the substrate within the enclosed region, the spacer being restrained by the frame structure; and

dispensing an underfill material within a dispense space that exists between the semiconductor die and the frame structure to form an underfill material layer, the underfill material layer being in contact with the peripheral walls of the frame structure and extending under the semiconductor die.

17. The method of forming the semiconductor assembly of claim **16**, wherein the frame structure is configured to prevent the underfill material layer from horizontally dispensing into the enclosed region.

18. The method of forming the semiconductor assembly of claim **16**, further comprising attaching a plurality of memory dies on the semiconductor die, the plurality of memory dies being at least partially carried by the spacer.

19. The method of forming the semiconductor assembly of claim **16**, wherein the spacer has a top planar surface.

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